Now, $[T(v)]_S$ is a column vector whose jth entry is, $a_{j1}k_1 + a_{j2}k_2 + \dots + a_{jn}k_n$

$\therefore$ jth entry of $[T]_S[v]_S$ = jth entry of $[T(v)]_S$

Hence $[T]_S[v]_S = [T(v)]_S$.

□

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$T: \mathbb{R}^2 \to \mathbb{R}^2$

Q:- $T(x,y) = (2x+3y, x-y)$

$S = \{(1,1), (1,2)\}$

$v = (2,3)$

$T(1,1) = (5,0) \Rightarrow [(5,0)]_S = (10,-5)^T$

$T(1,2) = (8,-1) \Rightarrow [(8,-1)]_S = (17,-9)^T$

$(a,b) = \alpha(1,1) + \beta(1,2)$

$= (\alpha+\beta, \alpha+2\beta)$

$\alpha + \beta = a$

$\alpha + 2\beta = b$

$\beta = b - a$

$\alpha = a - \beta$

$= a - b + a = 2a - b$

$$[T]_S = \begin{pmatrix} 10 & 17 \\ -5 & -9 \end{pmatrix}$$

$[(2,3)]_S = (1,1)^T$

$$[T]_S[v]_S = \begin{pmatrix} 10 & 17 \\ -5 & -9 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 27 \\ -14 \end{pmatrix}$$

$T(2,3) = (13,-1) \Rightarrow [(13,-1)]_S = (27,-14)^T$

$\therefore [T]_S[v]_S = [T(v)]_S$

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